Toward a four-dimensional Yang–Mills construction

Notebook 33: gauge-invariant determinant densities and Cartan probes

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Abstract

We identify a constant-curvature calculation with the actual Maxwell coefficient of the leading quantum correction. The starting point is the exact vector, based-ghost and effective-Hessian determinant identity of Notebook 32.

At a fixed composite endpoint, the matrices transform by background-gauge conjugation. Their diagonal color traces therefore define individually gauge-invariant analytic densities. Fixed-endpoint gaps and exponentially localized source kernels make their matrix logarithms local, uniformly in compatible volume. A rooted-density Ward argument proves that their quadratic affine coefficient is precisely the second-displacement moment of the total Hessian. This resolves the ambiguity that would arise from evaluating an arbitrary integration-by-parts density on an unbounded affine connection.

Commuting affine connections are exactly Wilson stationary and are preserved by the actual centered ordinary averaging maps. An exponential-kernel uniqueness argument identifies their polynomial stationary jets with the constructed minimum section. The envelope identity extends the zero-current conclusion to the effective classical actions. The constraint multiplier consequently vanishes through the background orders needed for the Maxwell calculation.

The resulting coefficient is an explicit diagonal trace expression on a constant Cartan-curvature probe, using the actual composite constraint and its normalized determinants. For the bare composite it involves the Wilson Hessian directly. The coarse contribution already has a uniform bound. We compute a specified massive continuum proper-time reference, including its positive logarithmic coefficient and a uniform finite-part bound. An exact heat-integral formula for the actual lattice coefficient isolates sufficient estimates on three heat-time intervals. Those estimates for the actual family remain incomplete. A closed-walk expansion proves the short-time bound for the actual based-ghost part, uniformly in endpoint and preceding depth. The vector part, the other heat-time intervals and the subsequent full-measure and continuum arguments remain.

All new arguments below are working proofs checked against the stated inputs from Notebooks 15–32. The matrix-logarithm locality estimates fix the endpoint before taking a volume limit. They are not estimates uniform in the ultraviolet depth. An affine field is used only through polynomial extensions of localized derivatives; no periodic family of continuously varying magnetic flux is assumed.

1 Gauge-invariant diagonal determinant densities

Fix a composite endpoint $K = \ell^k$, a preceding homogeneous depth M, and the actual finite periodic construction. The endpoint is fixed throughout this section. Constants may depend on K; they will be uniform in the permitted volume and in the preceding depth where the upstream classical bounds are. Choose the fixed sufficiently large scalar $a > 0$ from the coarse determinant estimate of Notebook 32.

At the actual stationary field $U_*(W)$, use its left-relative objects

$$C = f'(0), \quad s' = C^T\lambda, \quad \mathcal{L} = s'' - \lambda \cdot f'', \quad G_b = D_b^T D_b, \quad Q_b = \mathcal{L} + D_b D_b^T + aC^T C, \quad P_c = q_{\text{eff}} + aI.$$

The exact finite identity already established is

$$D_{M,K}(W) = \mathcal{N}_0 \left[\log\det G_b - \frac{1}{2}\log\det Q_b + \frac{1}{2}\log\det P_c \right]. \tag{1}$$

Here $\mathcal{N}_0$ subtracts the value at $W = 1$. The matrices act, respectively, on noncenter scalar sites, current bonds, and output bonds, including their color coordinates.

1.1 Background covariance of the actual matrices

For a fixed current-site gauge u, let O_E denote the block-diagonal adjoint action at each current bond's initial site, O_S the corresponding action at noncenter scalar sites, and O_C its action at output bond initial centers. These matrices are orthogonal for real gauges. They are complex orthogonal on the holomorphic continuation. The transformation of a left-relative variation is linear:

$$(e^{ix}U)^u = e^{iO_E x}U^u.$$

The actual ordinary average is equivariant, so its relative target logarithm transforms by O_C. Local gauge invariance of the two actual classical actions consequently gives

$$\begin{aligned} C(U^u) &= O_C C(U) O_E^T, & \lambda(W^{u|_{\text{centers}}}) &= O_C \lambda(W), \\ \mathcal{L}(U^u) &= O_E \mathcal{L}(U) O_E^T, & D_b(U^u) &= O_E D_b(U) O_S^T. \end{aligned} \tag{2}$$

For the Lagrangian identity, differentiate

$$s_{U^u}(O_E x) = s_U(x), \qquad f_{U^u}(O_E x) = O_C f_U(x).$$

The second equality and the multiplier transformation retain the entire $\lambda \cdot f''$ term. No unconstrained stationarity is assumed here.

The based scalar domain is unchanged by these transformations: zero values at all K-centers remain zero. The effective relative Hessian transforms by O_C, since its target relative variable also transforms linearly. Thus

$$G_b(U^u) = O_S G_b(U) O_S^T, \quad Q_b(U^u) = O_E Q_b(U) O_E^T, \quad P_c(W^{u|_{\text{centers}}}) = O_C P_c(W) O_C^T. \tag{3}$$

These identities concern the fixed scalar completion aI. They do not assert such covariance for an arbitrary background-dependent completion.

1.2 Fixed-endpoint locality of the matrix logarithms

We need locality of a diagonal logarithm, not merely locality of the trace of its derivatives. The required fixed-endpoint spectral and kernel inputs follow from the preceding modules, as follows.

At zero background, G_b has a positive gap independent of volume when K is fixed. Indeed, join every site to its block center by a path of length at most $c_d K$. For a scalar field vanishing at centers, Cauchy–Schwarz along these paths and summation in a block give

$$\|\phi\|_2^2 \le c_d K^{d+1} \|D_b \phi\|_2^2. \tag{4}$$

The crude exponent is sufficient here. This is compatible with the soft-mode upper bound in Notebook 32 and supplies no gap uniform as K grows.

The flat constrained contour precision has its previously proved positive gap at fixed K, uniformly in preceding depth and volume. On $\ker C$, decompose a vector into its component in $\operatorname{ran} D_b$ and its orthogonal complement. The form $\mathcal{L}$ annihilates the first. For a horizontal vector $v \in \ker C$, replace it

by $v + D_b T v$ to make its contour derivative zero. Its $\mathcal{L}$ value is unchanged, and the orthogonal projection of this replacement back to the horizontal space is v. The contour precision bound therefore controls $\|v\|_2^2$. Its passage from retained coordinates to current bonds uses the fixed-K bounded insertion K_0; this only changes the constant. On vertical vectors $D_b\phi$, the divergence penalty has the lower bound

$$\|D_b^T D_b \phi\|_2^2 \geq \lambda_{\min}(G_b)\|D_b\phi\|_2^2.$$

Thus $\mathcal{L} + D_b D_b^T$ has a uniform fixed-K gap on $\ker C$.

The actual adjusted minimum-section derivative

$$H_b = H - D_b G_b^{-1} D_b^T H$$

has $CH_b = I$, $D_b^T H_b = 0$, and a bounded operator norm at fixed K. The latter follows from the established minimum-response bound and the orthogonal projection in this formula. Write $x = \xi + H_b y$, $C\xi = 0$. The exact effective Schur identity gives

$$x^T Q_b x = \xi^T(\mathcal{L} + D_b D_b^T)\xi + y^T(q_{\text{eff}} + aI)y.$$

Since $q_{\text{eff}} + aI$ has a fixed positive gap, this proves a flat gap for Q_b, uniform in volume and preceding depth at fixed K. Their upper spectral bounds and exponentially weighted matrix bounds follow from the actual strict classical families, the finite-range constraint at fixed K, and the local gauge generator. P_c already has the stronger uniform bounds of Notebook 32.

Compose these matrices with the actual stationary section. Its strict support kernels, together with those of the classical actions and local charts, give source-rooted matrix derivative bounds through order two. A sufficiently small common complex output ball preserves the flat gaps in operator norm. This ball may depend on K, but not on volume or preceding depth.

Here is an elementary logarithm argument making the locality consequence precise. For any one of these matrix families $P(w)$, choose a fixed scalar s larger than its flat upper spectral bound and put

$$Z(w) = I - P(w)/s.$$

At zero $\|Z(0)\|_{2\to 2} < 1$, uniformly over the volumes under consideration. The exponentially weighted kernel bound implies that conjugating Z by a small exponential distance weight changes its operator norm by an arbitrarily small amount. First use bounded truncations of the distance weight; their uniform Lipschitz bound gives the same estimate, and then pass to kernel bounds. No unbounded similarity is assumed to act on all of ℓ^2. Shrink the background ball as well. The conjugated operators then have norm at most one fixed $r_K < 1$. Therefore

$$\log P(w) = (\log s)I - \sum_{n\geq 1} \frac{Z(w)^n}{n} \tag{5}$$

converges with these exponential weights. Termwise differentiation through order two is permitted: there are at most n or n^2 insertions, absorbed by the geometric factor r_K^{n-2}. The actual source-rooted bounds attach every background source to the matrix-product chain. Exponential weights then connect the diagonal matrix coordinate and all sources. Conversion from weighted ℓ^2 to row sums spends a smaller exponent and a finite geometric sum. The internal fiber per output block is finite at fixed K.

In particular, if a diagonal color trace is assigned to a fixed output block x, its quadratic kernel $k_x(b, c)$ obeys, for some $\kappa_K > 0$,

$$\sum_{b,c} e^{\kappa_K \tau(x,b,c)}|k_x(b, c)| \leq C_K. \tag{6}$$

Here the sum includes directions and colors, τ is a connecting-tree distance in output units, and all finite multiplicities of current coordinates are included. Periodic bounds use shortest lifted distances. No extensive logarithm is taken to an infinite-volume limit.

1.3 A density with its own gauge identity

Assign every noncenter current site to its unique centered K-block. Assign each positive current bond to the block containing its initial site, and each output bond to its initial output site. Define

$$\rho_x(w) = \mathcal{N}_0\Big[\sum_{\substack{z \text{ noncenter} \\ z \text{ in block } x}} \operatorname{tr}_{\mathfrak{g}}(\log G_b)_{zz} - \frac{1}{2} \sum_{\substack{e \text{ current} \\ e_- \text{ in block } x}} \operatorname{tr}_{\mathfrak{g}}(\log Q_b)_{ee} + \frac{1}{2} \sum_{\substack{c \text{ output} \\ c_- = x}} \operatorname{tr}_{\mathfrak{g}}(\log P_c)_{cc} \Big]. \tag{7}$$

All matrices in this formula are evaluated at $U_*(e^{iw})$ or its output e^{iw}, as appropriate. On the finite regulator,

$$D_{M,K}(e^{iw}) = \sum_x \rho_x(w) \tag{8}$$

exactly.

Functional calculus respects the similarities in (3). A diagonal color trace is unchanged by its site's orthogonal conjugation. Moreover, the actual minimum at gauge-transformed output is gauge equivalent to the transformed old minimum, on the established local germ. To see this, transform the constraint and action, then use the proved uniqueness of the local constrained critical orbit and its selected contour representative. The comparison need only hold on a finite real germ; analyticity supplies the local identities used below. It follows that *each* ρ_x, and each of its three displayed groups of terms, has its own local output gauge identity. This conclusion uses more than gauge invariance of their sum.

The block assignment is translation covariant. Local convergence of the matrices and their derivatives, together with (5) and its uniform geometric tail, passes these densities and their quadratic kernels to any jointly compatible infinite system at fixed endpoint. There they are translation covariant, satisfy (6), and retain the local gauge identity. No uniqueness of different structural limits is needed.

1.4 An affine-density formula for the total Maxwell moment

Lemma 1.1 (Gauge-invariant rooted density and total moment)**.** *Let ρ_x be translation-covariant analytic scalar densities on an infinite lattice, with exponentially rooted quadratic kernels as in (6). Assume every ρ_x is invariant under local gauge changes on its germ, and under global semisimple color conjugation. Fix one color and distinct directions μ, ν. For the formal affine component*

$$A_\mu(z) = t z_\nu, \qquad A_\sigma = 0 \quad (\sigma \neq \mu),$$

the quadratic density coefficient is

$$[t^2]\rho_0(A) = -\frac{1}{4} \sum_r r_\nu^2 K_{\mu\mu}(0, r), \tag{9}$$

where $K = \sum_x D^2\rho_x(0)$ is the total Hessian kernel in that color. Both sides are absolutely convergent kernel expressions. The left side does not mean evaluation of an analytic map at an unbounded nonzero field.

Proof. Global semisimple color invariance kills each density's linear derivative. Differentiating its local gauge identity at the origin therefore shows that its Hessian annihilates each compactly supported infinitesimal gradient in either slot. The exponential kernel bound extends this identity to polynomial

gauge parameters by absolute summation. In particular it annihilates a constant one-form in any one color and direction: use the linear scalar gauge parameter with that gradient.

Write $k_0(b,c)$ for the selected same-component Hessian. Its row and column sums are zero. Consequently

$$\frac{1}{2}\sum_{b,c} k_0(b,c) b_\nu c_\nu = -\frac{1}{4}\sum_{b,c} k_0(b,c)(c_\nu - b_\nu)^2.$$

Every interchange is allowed by exponential rooted summability. Translation covariance gives

$$K(0,r) = \sum_x k_x(0,r) = \sum_x k_0(-x, r-x).$$

The change of indices $b = -x,\ c = r - x$ turns the last second moment into the right side of (9). The same calculation with the translated affine field shows that the coefficient is independent of the selected density root. □

For the actual correction, the established Maxwell normalization is $\beta(D) = -\frac{1}{2}\sum_r r_\nu^2 K_{\mu\mu}(0,r)$, in one normalized color. Thus the constructed density gives

$$\boxed{\beta(D_{M,K}) = 2[t^2]\rho_0(t z_\nu e_\mu)}. \tag{10}$$

The density construction applies separately to the three groups of terms in (7), with the corresponding second-moment extraction on each. When the total has the established spatial symmetries, this is its Maxwell coefficient.

The individual gauge identity is essential. For example, the Maxwell action may be written either as its curvature square density or, after summation by parts, as a density of the form $\frac{1}{2}A \cdot d^* dA$. On an affine constant-curvature field the latter vanishes pointwise, while the former does not. Their totals agree on compact fields. The diagonal densities above avoid this ambiguity because their own gauge identities imply the row and column cancellations used in the proof.

1.5 Scope

The result identifies a specific, locally gauge-invariant trace density with the total Maxwell moment. It does not estimate that moment uniformly as K grows. The gap, analytic radius and decay exponent used for G_b, Q_b are permitted to deteriorate with K. No positive coefficient or comparison with a massive reference follows from the density identity alone.

2 Constant Cartan curvature and the actual stationary jets

We now use the diagonal-density construction to make a constant-curvature calculation an exact test of the actual Maxwell coefficient. All substitutions at an unbounded affine field mean substitutions into exponentially localized Taylor kernels at zero, through the required finite order.

2.1 Exact local Wilson and averaging identities

Choose commuting traceless Hermitian matrices $c_\mu, B_{\mu\nu}$ in a Cartan subalgebra of $\mathfrak{su}(N)$, with $B_{\mu\mu} = 0$, and put

$$A_\mu(x) = c_\mu + \sum_\nu B_{\mu\nu} x_\nu, \qquad U_\mu(x) = e^{iA_\mu(x)}. \tag{11}$$

The plaquette in the μ, ν plane is the constant matrix

$$e^{iF_{\mu\nu}}, \qquad F_{\mu\nu} = B_{\nu\mu} - B_{\mu\nu}.$$

The full Wilson Euler derivative is zero. For each varied bond and transverse direction, the two incident plaquettes give opposite transported sine terms. They agree after transport because all background links and plaquettes commute. The cancellation occurs as a Lie-algebra-valued identity, including against variations outside the Cartan subalgebra. This is a local infinite-lattice identity; an artificial finite-box boundary is not included.

For a centered primitive block, write $x = y + r$. Every coordinate-order root contour has transporter

$$\exp\{i[r \cdot A(y) + q_\Gamma(B, r)]\},$$

where q_Γ is independent of y. The selected mean is endpoint equivariant and conjugation equivariant. Its value on a tuple of diagonal matrices is diagonal: conjugation by every diagonal group element fixes the tuple, hence its output. Therefore the actual root transporter has the form

$$P_\ell(U; y, y + r) = e^{ir \cdot A(y)} p_\ell(B, r). \tag{12}$$

No log-arithmetic formula for the selected mean is assumed.

The p_ℓ factors cancel in the ordinary path product between neighboring blocks. Since $B_{\mu\mu} = 0$, this product is

$$U_x = e^{i\ell A_\mu(y)} \exp\left\{ i\ell \sum_\nu (B_{\mu\nu} - B_{\nu\mu}) r_\nu \right\}.$$

Its reference center string is $e^{i\ell A_\mu(y)}$. The defining ordinary average takes the centered average of the relative logarithms, and $\sum_r r_\nu = 0$. It follows that the actual primitive and iterated maps obey

$$M_\ell(U)(y, y + \ell e_\mu) = e^{i\ell A_\mu(y)}, \qquad F_K(U)(y, y + K e_\mu) = e^{iK A_\mu(y)}, \quad K = \ell^k. \tag{13}$$

Thus the output affine curvature is $K^2 F$. These are identities of local analytic germs, hence identities of formal curvature jets at each lattice site. They do not require a common bounded logarithm of the affine field on the whole infinite lattice.

Apply the actual direct-K root gauge $u(x) = P_K(U; y, x)$ in the block centered at y. It is the identity at centers. Endpoint covariance makes the transformed contour equal to the identity and leaves the composite output unchanged. Wilson stationarity is preserved by gauge invariance. At every fixed order in a curvature parameter, the rooted candidate has polynomial growth in the block center. The within-block coordinate set is finite at fixed K.

2.2 Why the formal candidate is the actual minimum jet

Lemma 2.1 (Polynomial extension of the stationary recursion)**.** *Suppose an actual stationary equation in retained coordinates is*

$$\mathcal{A}z + \mathcal{B}w + \mathcal{N}(w, z) = 0, \qquad \mathcal{N} = O((w, z)^2).$$

Assume $\mathcal{A}, \mathcal{A}^{-1}, \mathcal{B}$ have exponentially summable kernels, the inverse identities hold on finitely supported fields, and the mixed Taylor kernels of $\mathcal{N}$ through a specified finite order have exponential rooted-tree bounds. Then its analytic solution jets extend uniquely to polynomial-growth input jets. Any polynomial-growth formal solution has those same coefficients through that order.

Proof. For a polynomial-growth sequence v,

$$(1 + |j|)^m \le (1 + |i|)^m (1 + d(i, j))^m$$

shows that convolution by an exponentially summable kernel is absolutely defined and preserves polynomial growth. Exponential summability also permits the two convolutions in the inverse identities

to be interchanged. Those identities therefore hold on polynomial-growth sequences. In particular $\mathcal{A}$ has no nonzero polynomial-growth nullvector.

For a mixed kernel of any fixed finite degree, bound each input polynomial by a polynomial in the output coordinate times a polynomial in the joining-tree lengths. The exponential tree bound absorbs the latter factors. Thus the nonlinear Taylor coefficients are well defined. At order n the stationary equation is

$$\mathcal{A}z_n = h_n(w_1, \dots, w_n, z_1, \dots, z_{n-1}),$$

and has the unique polynomial-growth solution $\mathcal{A}^{-1}h_n$. These are also the equations obtained by differentiating the analytic bounded-field solution. Induction proves the assertion. □

For the actual constrained stationary branch at a fixed endpoint, these are established inputs: the flat retained precision has its exponential inverse, the affine constraint chart is local, and the classical scalar and stationary support families have exponentially localized mixed derivatives. The finite fiber dimension is per block; the whole lattice is not finite-dimensional in this application. Only finite-order jets are being extended.

For the bare Wilson action, the rooted affine candidate above solves the constraint, contour and stationary equations coefficientwise, with unconstrained current and multiplier equal to zero. Lemma 2.1 therefore identifies it with the actual stationary section's polynomial jets. In particular this applies through the two background orders needed below.

2.3 The effective classical action is also stationary on this family

The effective-action interface can be checked using the actual envelope identity, without replacing its Hessian by a bare Wilson Hessian.

Fix any permitted finite homogeneous depth M. The action S_M is the actual local minimum value of the bare Wilson action under F_M. Its differential in left-relative output coordinates equals the target multiplier of that minimum. This identity is an equality of the established analytic kernels. It extends to polynomial input jets by their exponential bounds.

Apply the preceding bare-action argument at the fixed endpoint M, with fine curvature equal to the prescribed current curvature divided by M^2. The explicit fine candidate is Wilson stationary, so its multiplier jets are zero. Consequently

$$D_{\mathrm{left}} S_M(U^{\mathrm{affine}}(t)) = 0 \quad \text{as formal jets through order two.} \tag{14}$$

The identity holds in every current-link component; it is not just a derivative along the one-parameter family. Indeed the envelope identifies the whole target multiplier, and the unconstrained fine Euler derivative is zero.

At the next fixed endpoint K, take the current affine field with curvature divided by K^2, and root it by the actual direct-K based gauge. The classical local gauge identity, also extended coefficientwise by exponential bounds, preserves (14). This candidate satisfies the actual S_M constrained stationary equation with zero multiplier through the same orders. Apply Lemma 2.1 to the retained precision for S_M. The candidate has exactly the actual current stationary jets, and

$$\lambda_{M,K}(t) = O(t^3) \tag{15}$$

in this formal sense.

All assertions so far fix each endpoint when invoking its inverse. They do not infer a gap uniform in M or K for a one-shot fine operator. However, the current-coordinate classical and stationary Taylor kernels at fixed K have the already established uniform bounds in preceding depth. Their polynomial moments consequently pass to any varying-depth structural limit by exponential domination. Thus (14) and (15) also hold in those joint systems. The effective Hessian s'' still depends on S_M; this result does not make it the bare Wilson Hessian.

2.4 An exact constant-curvature expression for the Maxwell coefficient

Fix a trace-orthonormal Cartan color H, and distinct directions μ, ν. The output probe is

$$w_\mu(n;t) = tn_\nu H, \qquad w_\sigma = 0 \quad (\sigma \neq \mu).$$

Its unrooted current affine representative is

$$U_\mu^{\mathrm{aff}}(x;t) = \exp(itx_\nu H/K^2), \qquad U_\sigma^{\mathrm{aff}} = I \quad (\sigma \neq \mu). \tag{16}$$

By (13), its output is exactly the chosen formal probe. Its oriented μ, ν plaquette has angle $-tH/K^2$.

On this family write

$$\begin{aligned} G(t) &= D_b(U^{\mathrm{aff}}(t))^T D_b(U^{\mathrm{aff}}(t)), \\ V_a(t) &= s''(U^{\mathrm{aff}}(t)) + D_b(U^{\mathrm{aff}}(t)) D_b(U^{\mathrm{aff}}(t))^T + aC(U^{\mathrm{aff}}(t))^T C(U^{\mathrm{aff}}(t)), \\ P(t) &= q_{\mathrm{eff}}(w(t)) + aI. \end{aligned} \tag{17}$$

All are formal matrix jets around their flat positive operators. In particular no invertibility or positivity on a nonzero global magnetic background is assumed. Equation (15) removes the $-\lambda \cdot f''$ term through second order. The remaining Hessian s'' retains the actual action.

Use the same block-zero site and bond assignments as in the diagonal-density construction. Its pointwise gauge invariance allows removal of the root gauge, coefficientwise: the local gauge identities extend to these polynomial gauge jets by the same exponential summation. The density–moment lemma therefore gives

$$\begin{aligned} \beta(D_{M,K}) = [t^2]\Big\{ & 2 \sum_{\substack{z \text{ noncenter} \\ z \text{ in block } 0}} \mathrm{tr}_{\mathfrak{g}}(\log G(t))_{zz} \\ & - \sum_{\substack{e \text{ current} \\ e_- \text{ in block } 0}} \mathrm{tr}_{\mathfrak{g}}(\log V_a(t))_{ee} \\ & + \sum_{\substack{c \text{ output} \\ c_- = 0}} \mathrm{tr}_{\mathfrak{g}}(\log P(t))_{cc} \Big\}. \end{aligned} \tag{18}$$

Field-independent subtractions can be omitted inside $[t^2]$. The factors $2, -1, +1$ follow from $\beta = 2[t^2]\rho_0$, and the determinant weights $1, -\frac{1}{2}, +\frac{1}{2}$.

Every coefficient on the right is defined by the localized matrix-logarithm derivatives at zero. Their source kernels absorb the polynomial weights, so the sums are convergent at fixed endpoint. For $M = 1$, s'' in $V_a(t)$ is the actual bare Wilson Hessian on an explicit commuting constant-curvature background. The coarse contribution is uniformly bounded in M, K by Notebook 32. Hence the remaining ultraviolet comparison can be made directly on the first two terms of (18).

2.5 What this settles and what it does not

The formula matches an explicit constant-curvature trace calculation to the actual Maxwell extraction, with the actual composite average, stationary jets and measure factors. It removes the multiplier-curvature jets on this probe. It does not replace an arbitrary local density by its value on an affine field: the individual gauge identity of the chosen diagonal densities was essential.

On a fixed periodic torus, constant nonzero Cartan flux has quantization conditions. No analytic family of periodic fluxes through zero is used here. Nor is an unbounded affine field substituted into the original bounded-field map. The identities are polynomial extensions of localized derivatives in an infinite joint system.

The fine vector–ghost expression still needs a bound on its difference from a precisely normalized reference logarithm, uniform as K grows. The fixed-endpoint logarithm construction supplies no such bound and computes no sign. Mixed and large-field histories, the continuum observable construction and reconstruction remain separate requirements for full existence.

3 A completely specified massive continuum reference

We compute a reference with its regulator and normalization fixed. This supplies a target for comparison, not an identification of the actual constrained lattice determinant with that target. The background-field operators and heat coefficient agree with the conventions described in [1], Sections 3.4, 4.2.1 and 8.3. The calculation below keeps the regulator and the one-color Maxwell extraction explicit.

Use final physical coordinates on $\mathbb{R}^4$, scalar mass $m > 0$, and a lower proper-time cutoff $\epsilon^2 > 0$. Choose a Hermitian Cartan element H with $\tau(H^2) = 1$. Let q run over the eigenvalues of ad_H on the complexified adjoint space, with multiplicities. Both signs of a nonzero charge are included. Complexification has the same trace as the original real color space; it does not introduce another factor of two. Put

$$C_H = \sum_q q^2.$$

For one constant curvature component $F_{12} = fH$, a charge sector has scalar operator $L_q = -D_\mu D_\mu$, where one may choose $D_2 = \partial_2 + iqfx_1$ and the other derivatives ordinary. The vector reference is

$$(V_q)_{\mu\nu} = L_q\delta_{\mu\nu} - 2iqF^{(1)}_{\mu\nu}, \qquad F^{(1)}_{12} = f, \quad F^{(1)}_{21} = -f.$$

The spin matrix has eigenvalues $0, 0, 2qf, -2qf$. Its sign convention has no effect on its trace. We add m^2 to both operators.

3.1 Exact diagonal heat traces

Fourier transformation in x_2 reduces the magnetic plane to oscillators with eigenvalues $|qf|(2n+1)$. For a normalized oscillator eigenfunction, integration of its shifted squared value gives

$$\int_{\mathbb{R}} \frac{dp_2}{2\pi} \left|\psi_n\Big(x_1 + \frac{p_2}{qf}\Big)\right|^2 = \frac{|qf|}{2\pi}.$$

Thus this is a pointwise diagonal density calculation, without a finite-box degeneracy assumption. Summing the oscillator levels and multiplying by the heat density of the two free coordinates gives

$$k_q(s; f) = (4\pi s)^{-2} \frac{qfs}{\sinh(qfs)}. \tag{19}$$

At $qf = 0$ the fraction is defined by its limit. The vector density is

$$k_q(s; f)\{2 + 2\cosh(2qfs)\}.$$

Define the normalized proper-time determinant correction with weights $-\frac{1}{2}\log\det(V + m^2) + \log\det(L + m^2)$. Here this phrase means the following density integral; there is no infinite-volume determinant:

$$\mathcal{R}_{\epsilon,m}(f) = \frac{1}{16\pi^2} \int_{\epsilon^2}^{\infty} \frac{ds}{s^3} e^{-m^2 s} \sum_q \left\{ \frac{qfs}{\sinh(qfs)} \cosh(2qfs) - 1 \right\}. \tag{20}$$

The sign follows from $\log P = -\int ds\, e^{-sP}/s$, after subtraction and the specified proper-time cutoff. The two vector polarizations with zero spin cancel the scalar ghost in this combined expression.

For real $|f|\max_q|q| < m^2$, the massive vector has positive bottom and the integral converges at infinity. On a smaller real interval, differentiation twice at zero is justified by a common exponentially decaying majorant. In particular, the notation $[f^2]$ below denotes half the second real derivative at zero. We do not assert a convergent complex Taylor series in a nonzero constant magnetic field.

Proposition 3.1 (Normalized reference coefficient)**.** *Set $E_1(x) = \int_x^\infty e^{-u}\,du/u$. In the normalization $\beta = 2[f^2]$ used in this notebook,*

$$\beta_{\mathrm{ref}}(\epsilon, m) = \frac{11C_H}{48\pi^2} E_1(m^2\epsilon^2). \tag{21}$$

In particular, with $m = 1$, $\epsilon = K^{-1}$,

$$\left|\beta_{\mathrm{ref}}(K^{-1}, 1) - \frac{11C_H}{24\pi^2}\log K\right| \le \frac{11C_H}{48\pi^2}(1 + e^{-1}), \qquad K \ge 1. \tag{22}$$

Proof. The scalar magnetic factor and the spin factor give

$$\frac{x}{\sinh x}\cosh(2x) = 1 + \frac{11}{6}x^2 + O(x^4).$$

Differentiating (20) twice at zero therefore gives

$$[f^2]\mathcal{R}_{\epsilon,m} = \frac{11C_H}{96\pi^2}\int_{\epsilon^2}^\infty \frac{e^{-m^2 s}}{s}\,ds.$$

The substitution $u = m^2 s$ proves (21). For $0 < x \le 1$,

$$E_1(x) - \log(1/x) = \int_x^1 \frac{e^{-u} - 1}{u}\,du + \int_1^\infty \frac{e^{-u}}{u}\,du.$$

The absolute value is at most $1 + e^{-1}$. This proves the asserted uniform finite part. □

For the repository's $SU(N)$ normalization $\tau = \mathrm{Tr}_{\mathrm{fund}}/N$, write $H = \mathrm{diag}(h_1, \dots, h_N)$. Then $\sum_i h_i = 0$, $\sum_i h_i^2 = N$, and the nonzero-color charges are $h_i - h_j$, $i \ne j$; the $N - 1$ Cartan charges vanish. Consequently

$$C_H = \sum_{i,j}(h_i - h_j)^2 = 2N\sum_i h_i^2 = 2N^2.$$

The coefficient of $\log K$ is thus $11N^2/(12\pi^2)$, or $11/(3\pi^2)$ for $SU(2)$, in this one-normalized-color convention. This is not a sum of $N^2 - 1$ different Maxwell extractions.

3.2 Scale and regulator bookkeeping

The same continuum reference can be written in current fine units, with cutoff $s \ge 1$, mass K^{-1}, and curvature f/K^2. Multiplication of that density by the block volume K^4, followed by $s = K^2 u$, gives exactly (20) with $\epsilon = K^{-1}$, $m = 1$. Thus there is no additional factor of K^4 in its extracted coefficient. This is a change of units within the continuum reference, not a replacement of a lattice heat kernel by a continuum kernel.

Fixed positive changes of these continuum regulators have a bounded effect. For $c, m > 0$,

$$|\beta_{\mathrm{ref}}(c/K, m) - \beta_{\mathrm{ref}}(1/K, 1)| \le \frac{11C_H}{24\pi^2}|\log(cm)|. \tag{23}$$

Indeed, the difference of the two E_1 integrals is bounded by the integral of du/u between their lower limits. Uniformity follows when cm remains in a fixed compact subinterval of $(0, \infty)$. This estimate does not compare a based ghost to an unbased massive ghost, change a lattice constraint, or control a massless zero-mode limit.

4 The actual heat integral and a sufficient comparison estimate

The Cartan identity also puts the missing comparison into heat-kernel form without changing the lattice operators. Use $G(t), V_a(t)$ from (17). Let $\mathrm{Tr}_{0,S}$ be the diagonal color trace over noncenter scalar sites in block zero and $\mathrm{Tr}_{0,E}$ that over current bonds with initial site in block zero. Define the formal coefficient

$$h_{M,K}(s) = [t^2]\left\{\mathrm{Tr}_{0,E}\, e^{-sV_a(t)} - 2\,\mathrm{Tr}_{0,S}\, e^{-sG(t)}\right\}. \tag{24}$$

Every derivative here is evaluated at the flat positive operators and on the polynomial stationary jets. No heat evolution at a nonzero unbounded background is used. Put

$$b_c(M,K) = [t^2]\sum_{c_-=0} \mathrm{tr}_{\mathfrak{g}}(\log P(t))_{cc}.$$

Notebook 32 bounds $|b_c|$ uniformly in M, K.

Proposition 4.1 (Exact actual heat representation)**.** *In each joint infinite system at fixed K,*

$$\beta(D_{M,K}) = b_c(M,K) + \int_0^\infty h_{M,K}(s)\,\frac{ds}{s}. \tag{25}$$

The integral is absolutely convergent for fixed K. The bounds establishing this statement may depend on K.

Proof. First use bounded compactly supported output directions. For a positive flat matrix family $P(t)$, the local functional calculus gives

$$[t^2](\log P(t))_{ii} = -\int_0^\infty [t^2](e^{-sP(t)})_{ii}\,\frac{ds}{s}.$$

One can obtain this by differentiating the difference of the proper-time formulas for $P(t)$ and $P(0)$; their scalar reference term cancels. To justify the same formula on polynomial jets, write $P(t) = P_0 + tP_1 + t^2P_2 + O(t^3)$. Its heat coefficient is

$$\begin{aligned}[t^2]e^{-sP(t)} = &-\int_0^s e^{-(s-u)P_0}P_2e^{-uP_0}\,du \\ &+\int_{0<u<v<s} e^{-(s-v)P_0}P_1e^{-(v-u)P_0}P_1e^{-uP_0}\,du\,dv.\end{aligned} \tag{26}$$

The P_j kernels are polynomially growing at worst, with exponential localization in their relative coordinates. This follows by inserting the affine source jets in the established rooted derivative kernels.

The fixed-K flat gap and exponential matrix bound give exponentially weighted semigroup estimates. Indeed bounded truncated distance conjugation changes P_0 in operator norm by less than half its gap if the exponent is small enough. The variation-of-constants bound then gives $\|e^{-sP_0}\|_{\text{weighted}} \le e^{-c_K s}$. Spending smaller spatial exponents absorbs the polynomial factors in (26), after anchoring at the diagonal block. Its trace is bounded by

$$C_K(s+s^2)e^{-c'_K s}, \qquad s>0,$$

with constants uniform in volume and, where the upstream bounds are, preceding depth. For small s the same estimate follows directly from the one and two insertion formulas. These bounds justify the proper-time integral, polynomial-jet extension and local infinite-volume passage by domination. Substituting the two log coefficients in (18) proves (25), with signs $+1, -2$ in (24). □

4.1 A uniform short-time bound for the actual based ghost

One part of the short-time comparison can be proved directly. It uses the actual based scalar domain and has constants independent of the preceding depth and endpoint.

Proposition 4.2 (Actual ghost ultraviolet bound)**.** *In four dimensions put*

$$h_K^g(s) = -2[t^2]\,\mathrm{Tr}_{0,S}\, e^{-sG(t)}.$$

For every permitted K, on the affine probe already identified with the actual stationary jets,

$$0 \le h_K^g(s) \le C_H\{(8s)^4 + 6(8s)^3 + 7(8s)^2 + 8s\}, \qquad s > 0. \tag{27}$$

Consequently

$$\int_0^1 |h_K^g(s)|\,\frac{ds}{s} \le 2280 C_H. \tag{28}$$

Equivalently, the based-ghost part of the rescaled short-time integral over $0 < u < K^{-2}$ is bounded by this same constant.

Proof. On the complexified charge-q space, the actual based operator is the covariant nearest-neighbor Laplacian with Dirichlet values at every K-center. Hence

$$G_q(t) = 8I - A_q(t),$$

where A_q is the covariant adjacency operator on noncenter sites. The diagonal is still 8, also at sites adjacent to a removed center: it comes from the full link norm of D_b, with the center value set to zero.

The diagonal of $A_q(t)^n$ is a sum over length-n closed nearest-neighbor walks avoiding the centers. On the commuting affine field each walk has phase

$$\exp\{iqt\,\mathcal{A}_{\mu\nu}(\gamma)/K^2\},$$

where $\mathcal{A}_{\mu\nu}(\gamma)$ is its oriented algebraic area in the selected coordinate plane. The contribution from a constant potential cancels on a closed walk. Taking the origin of the area coordinate at the starting site gives the elementary bound $|\mathcal{A}_{\mu\nu}(\gamma)| \le n^2$. Thus the block trace coefficient is exactly

$$h_K^g(s) = \frac{C_H}{K^4} e^{-8s} \sum_{\substack{z \text{ noncenter} \\ z \text{ in block } 0}} \sum_{n\ge 0} \frac{s^n}{n!} \sum_{\substack{\gamma: z\to z,\ |\gamma|=n \\ \gamma \text{ avoids centers}}} \mathcal{A}_{\mu\nu}(\gamma)^2. \tag{29}$$

All sums converge absolutely at each fixed s; the factorial absorbs the factor n^4. This also justifies extracting the formal quadratic coefficient without imposing periodic magnetic flux. Every term is nonnegative.

There are at most K^4 starting sites and at most 8^n walks from each. It follows that

$$h_K^g(s) \le C_H e^{-8s} \sum_{n\ge 0} \frac{(8s)^n n^4}{n!}.$$

The fourth moment of this exponential generating series is $x^4 + 6x^3 + 7x^2 + x$ after multiplication by e^{-x}; it follows by applying $(x\,d/dx)^4$ to e^x. This proves (27). Integrating its four monomials from zero to one with measure ds/s gives

$$C_H\left(\frac{8^4}{4} + \frac{6\,8^3}{3} + \frac{7\,8^2}{2} + 8\right) = 2280 C_H.$$

□

This bound is uniform in M because the actual stationary ghost jets on this probe have already been identified with the same affine covariant Laplacian for every S_M. No estimate for a separate massive inverse is used. The polynomial upper bound in (27) is not a useful large-time bound and does not settle the long-time comparison.

4.2 A quantitative target in physical heat time

Set $J_{M,K}(u) = h_{M,K}(K^2 u)$, and $\alpha_H = 11C_H/(48\pi^2)$. The continuum calculation above has reference heat coefficient

$$J_K^{\mathrm{ref}}(u) = \alpha_H e^{-u} \mathbf{1}_{\{u \geq K^{-2}\}}.$$

Thus the following is an exact difference identity, not yet a bound:

$$\beta(D_{M,K}) - \beta_{\mathrm{ref}}(K^{-1}, 1) = b_c(M,K) + \int_0^\infty \{J_{M,K}(u) - J_K^{\mathrm{ref}}(u)\} \frac{du}{u}. \tag{30}$$

It fixes the actual operator, regulator, sign, density, and rescaling in a proposed comparison.

Lemma 4.3 (Sufficient bounds on three heat-time intervals)**.** *Suppose, for all required M, K and structural systems, there are constants $C, \delta > 0$, independent of them, such that*

$$\int_0^{K^{-2}} |J_{M,K}(u)| \frac{du}{u} \leq C, \tag{31}$$

$$|J_{M,K}(u) - \alpha_H| \leq C\{(K^2 u)^{-\delta} + u^{\delta}\}, \quad K^{-2} \leq u \leq 1, \tag{32}$$

$$\int_1^\infty |J_{M,K}(u)| \frac{du}{u} \leq C. \tag{33}$$

Then, with the already uniform coarse bound C_c,

$$\left| \beta(D_{M,K}) - \frac{11C_H}{24\pi^2} \log K \right| \leq C_c + 2C + \frac{2C}{\delta}. \tag{34}$$

Proof. Integrate the exact identity (25) after the time rescaling. The constant α_H on the middle interval integrates to $2\alpha_H \log K$. Each error in (32) integrates to at most C/δ. The other intervals and the coarse term give the remaining terms. □

The full bounds (31)–(33) are not asserted here for the actual family uniformly in K. Proposition 4.2 supplies the ghost part of the first bound; its vector part remains. The fixed-endpoint heat representation by itself gives only constants depending on K. The middle interval asks for a continuum coefficient on heat lengths between the fine spacing and the final block size. The last interval is where the soft pinned-ghost modes are particularly relevant: the scalar gap alone may be of order K^{-4} in fine units, so rescaling by K^2 does not give uniform damping. The combined coefficient must be estimated, or a weaker signed integral comparison must be proved. Bounding the separate inverse operators by a physical mass is not an available shortcut.

For $M > 1$, V_a contains the actual S_M Hessian. Uniformity over that family is part of the conditions, not a consequence of computing the bare Wilson reference. A bounded comparison only at $M = 1$ would not by itself give the varying-depth margin needed by the finite drift-transfer theorem.

5 The remaining scale-dependent comparison

For the bare composite, the new identification reduces the leading coefficient, up to the uniformly bounded coarse effective-Hessian term, to the constant-curvature jets of

$$\log\det(D_b^T D_b) - \frac{1}{2}\log\det(S_{\mathrm{W}}'' + D_b D_b^T + a C_K^T C_K).$$

This is a statement about the actual diagonal trace density and its Maxwell moment. The formal Cartan probe makes the constrained-current curvature vanish in the required jets. It does not make either determinant individually uniformly massive at the final scale.

The next estimate must compare this combined expression to a fully specified background-Feynman vector and ghost reference with infrared scale K^{-1}, preserving its coarse-coordinate, color and action normalization. In particular, changing the based scalar domain, the vector constraint completion and the infrared regulator must be controlled together. The based-ghost Rayleigh bound from Notebook 32 still rules out a separate inverse bound of order K^2.

A sufficient result would give a constant C, independent of K, bounding the difference of the extracted Maxwell coefficients. The specified continuum reference now has an explicit positive logarithmic law and bounded finite part. The exact actual heat representation identifies the remaining comparison integral; the three interval estimates in Lemma 4.3 would bound it. Those estimates do not follow from the fixed-K convergence used here. The actual based-ghost part of the short-time interval is now bounded uniformly by the closed-walk argument; its vector counterpart and the other intervals remain. Uniformity over the structural systems needed by the finite drift-transfer theorem must also be retained.

Once positive averaged drift is established with that scope, the conditional finite shooting theorem applies. The full four-dimensional existence objective still requires the other field sectors, the exact global measure representation, sourced continuum bounds, nontriviality, Euclidean symmetry, positivity and reconstruction.

References

[1] D. V. Vassilevich, *Heat kernel expansion: user's manual*, Physics Reports 388 (2003), 279–360, arXiv:hep-th/0306138. Sections 3.4, 4.2.1 and 8.3 specify the background vector and ghost operators, their combined heat coefficient, and the constant magnetic scalar heat kernel. This source is not used as a proof of the actual constrained lattice comparison.

[2] *Toward a four-dimensional Yang–Mills construction*, research Notebooks 15–32 in this workspace. The actual local minimum, strict mixed kernels, classical envelope, constrained Gaussian, gauge identities, determinant bridge and Maxwell normalization are the inputs used here.

[3] E. Sussman and collaborators, Yang–Mills-LLM-experiment, pinned deposited accounts at commit `e392ef45013a581664bab76b90867170b30fe10e`. The centered ordinary average and its selected endpoint-equivariant mean are kept unchanged.